
APEX QUANT: A MULTI-AGENT DEBATE FRAMEWORK FOR QUANTITATIVE TRADING

TECHNICAL REPORT

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March 30, 2026

ABSTRACT

We present Apex Quant, a multi-agent large language model (LLM) framework for quantitative trading decisions. The system employs three agents with distinct personas—an aggressive bull (Zealot), an aggressive bear (Reaper), and a moderating chief investment officer (Fulcrum)—engaged in structured multi-round debate before reaching a trading decision. Rather than reporting aggregate performance statistics, this paper documents a series of non-obvious architectural discoveries emerging from iterative development and live forward testing: that debate structure itself provides decision stability rather than prompt memory; that systematic persona bias can emerge wherein one agent consistently dominates through rhetorical structure rather than evidence quality; and that distilling behavioral finance principles directly into agent prompts produces qualitatively superior reasoning compared to relying on base model priors. We describe the full evolution of the system, including failed architectural variants, and present case studies from live forward testing that illustrate both the system’s decision mechanisms and the structural tensions inherent in its design—serving as a practical guide for researchers building LLM-based trading agents. The system is publicly available at https://github.com/sst19910323/apex_parliament.

Keywords Large language models · Multi-agent systems · Quantitative trading · Debate architecture · Behavioral finance · Investment decision making

1 Introduction

Quantitative trading has long relied on statistical models, rule-based systems, and more recently, machine learning approaches to extract signal from financial markets. These methods share a common limitation: they optimize for pattern recognition within historical data, with limited capacity to reason about novel market conditions or integrate qualitative judgment at the moment of decision.

Large language models (LLMs) offer a fundamentally different capability: the ability to reason over heterogeneous inputs, weigh competing hypotheses, and articulate conditional judgments in natural language [Wei et al., 2022]. However, deploying a single LLM as a trading agent introduces well-documented failure modes—overconfidence, susceptibility to confirmation bias, and inconsistent behavior across sessions [Sharma et al., 2023, Gallegos et al., 2024].

We propose that these failure modes are not primarily prompt engineering problems, but architectural ones. A single agent, regardless of how carefully its prompt is designed, lacks an internal mechanism for adversarial self-correction. Our solution is a structured three-agent debate framework in which agents with fixed, opposing investment philosophies are required to build independent cases before deliberating toward a consensus position. Concurrently and independently, Xiao et al. [2024] arrived at a structurally similar conclusion through a different design philosophy, providing convergent evidence that adversarial debate addresses a genuine architectural gap in LLM-based trading.

The resulting system, Apex Quant, has been deployed in live forward testing on Chinese A-share and futures markets. More importantly for the research community, the development process surfaced several architectural insights that we believe generalize beyond this specific application.

The primary contributions of this paper are as follows:

- A three-agent debate architecture for trading decisions, with a principled rationale for persona design and debate protocol structure.
- The identification and partial resolution of *systematic persona bias*: a phenomenon where one agent consistently wins debates through rhetorical framing rather than evidence quality.
- Empirical evidence that distilling behavioral finance principles into agent system prompts—rather than relying on emergent reasoning—produces measurable improvements in Sharpe ratio.
- A documented development history including failed architectural variants, offered as practical guidance for future work.

The remainder of this paper is organized as follows. Section 2 reviews related work on LLM agents and algorithmic trading. Section 3 describes the system architecture. Section 4 presents key design principles derived from iterative development. Section 5 documents the broader exploration process, including failed architectural variants. Section 6 presents case studies from live forward testing and backtesting, illustrating representative trading decisions alongside two cases of particular structural and analytical interest. Section 7 concludes.

2 Related Work

2.1 LLM Agents for Financial Decision Making

The application of large language models to financial tasks has expanded rapidly. Early work demonstrated that LLMs possess substantial financial knowledge and can perform sentiment analysis, earnings call summarization, and basic market commentary [Lopez-Lira and Tang, 2023, ?]. More recent efforts have explored LLM-based trading agents that consume market data and generate buy/sell signals directly [??].

However, single-agent approaches face a fundamental tension: the same model that reasons fluently about market conditions is also prone to overconfident position-taking, anchoring to recent context, and inconsistent risk tolerance across sessions. ? document systematic hallucination in financial LLM applications, where models generate plausible but factually incorrect reasoning chains. Our architecture addresses this by structuring disagreement rather than attempting to eliminate it.

2.2 Multi-Agent Debate and Deliberation

The use of multiple LLM agents to improve reasoning quality has gained significant attention. Du et al. [2023] show that having multiple LLM instances debate a question and update their answers based on responses from other agents leads to substantially improved factual accuracy and mathematical reasoning. Liang et al. [2023] demonstrate that agent disagreement, when properly structured, reduces sycophancy and improves output diversity.

A critical design distinction in our work is between *refutation-based* and *constructive* debate architectures. In refutation-based designs, agents primarily respond to and attack each other’s positions. We find this leads to rhetorical dominance effects rather than evidence-based consensus. Our constructive design requires each agent to build an independent case before deliberation begins, a distinction elaborated in Section 3.

2.3 Behavioral Finance and Systematic Bias

Behavioral finance research has catalogued a rich set of cognitive biases that affect human traders: loss aversion [Kahneman and Tversky, 1979], momentum chasing, and overreaction to recent events [?]. A non-obvious finding of our work is that LLM agents reproduce several of these biases spontaneously—not because they are explicitly prompted to, but because the biases are embedded in the training corpora derived from human-generated financial text. This motivates our approach of explicitly encoding counter-bias principles into agent system prompts, described in Section 4.

2.4 Quantitative Trading Systems

Classical quantitative approaches range from statistical arbitrage [?] to deep reinforcement learning [??]. While these methods excel at exploiting well-defined statistical regularities, they require large labeled datasets and struggle

with regime changes. Our system occupies a complementary niche: it is designed for low-frequency, high-conviction decisions where qualitative reasoning about market context is as important as quantitative signal.

3 System Architecture

3.1 Design Philosophy

A central premise of Apex Quant is that trading decision quality is fundamentally a problem of structured disagreement, not information aggregation. A single agent given all available market data will tend to converge on a dominant narrative—whichever interpretation is most salient in its training distribution. Real investment committees derive value not from averaging opinions, but from forcing independent, opposing cases to be articulated and defended before a decision is reached.

This observation motivates a three-agent round-table architecture in which no agent is designed to be neutral or balanced. Each agent holds a fixed investment philosophy that it is constitutionally required to defend. Stability emerges from the tension between agents, not from any individual agent’s internal calibration. All three agents participate as equals; Fulcrum additionally serves as facilitator and rapporteur, a practical division of labor rather than a hierarchy of authority.

3.2 Agent Personas

The three agents and their roles are illustrated in Figure 1. We name them Zealot, Reaper, and Fulcrum.

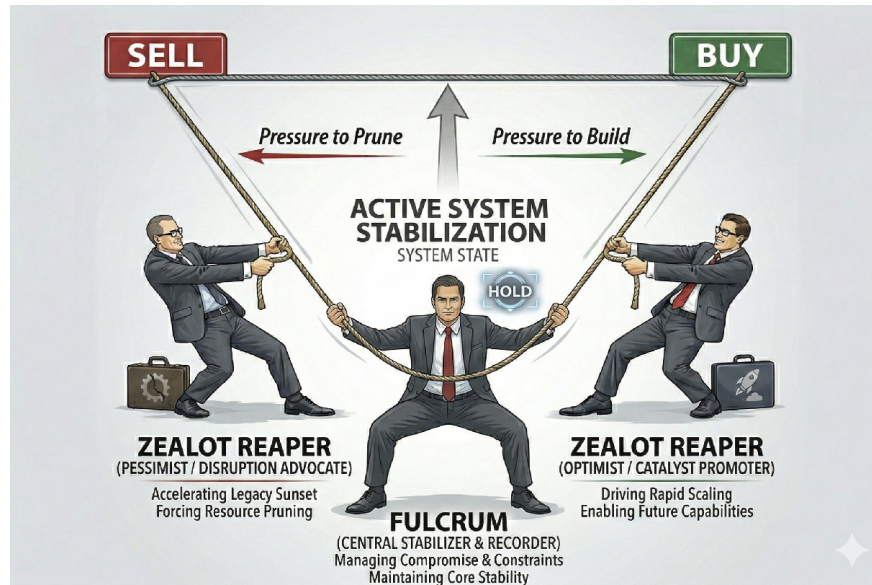


Figure 1: The three-agent round-table architecture of Apex Quant. Zealot (Analyst) pulls toward BUY; Reaper (Auditor / Cash Holder) pulls toward SELL; Fulcrum (CIO) holds the center. The system’s equilibrium is a conservative HOLD bias, which shifts only under the weight of well-evidenced arguments from either side.

Zealot occupies the aggressive analyst role. Its mandate is to identify actionable long opportunities in current market conditions and to construct the strongest possible case for position entry. Zealot is explicitly instructed to seek asymmetric upside, to argue that the current thesis remains intact, and to resist premature exit. It is not a balanced analyst: it is constitutionally bullish, and its value to the debate lies precisely in its refusal to hedge. Zealot’s characteristic failure mode—which the architecture is designed to contain rather than eliminate—is the tendency to extend a valid thesis beyond its natural expiration, finding reasons to stay in a position after the original catalyst has been exhausted.

Reaper occupies the capital preservation and profit-taking role. Its deeper mandate is realism: to ask, at every decision point, whether remaining in a position is still justified by current evidence, or whether the prudent action is to harvest existing gains and return to cash. Where Zealot argues that the thesis remains intact and further upside lies ahead, Reaper argues that the thesis has been fulfilled, the easy money has been made, and the risk/reward of continued holding no longer justifies the exposure. In the majority of cases, Reaper’s exit argument is not “the market is collapsing”—that

scenario is relatively rare—but the quieter and more common case: “we have made a reasonable return, conditions are uncertain, and cash is a position.” The name reflects not an appetite for decline, but a willingness to harvest gains that Zealot would leave on the table in pursuit of further upside.

Fulcrum participates in the debate as an equal, but additionally serves two procedural roles: maintaining debate order and authoring the final report. Fulcrum is conservative by disposition—its natural tendency is to pull both agents toward the center, and its default output in the absence of compelling evidence is **HOLD**. This conservatism is a persona characteristic, not a structural privilege; Fulcrum’s vote carries no more weight than Zealot’s or Reaper’s during deliberation. Its role as rapporteur is a division of labor: someone must synthesize and record, and the most conservative agent is the natural choice for a task that rewards caution over advocacy.

A key design principle is that all three agents first build *independent* assessments before seeing each other’s outputs. This prevents the debate from collapsing into refutation loops where rhetorical skill dominates evidence quality.

3.3 Debate Protocol

Each decision cycle proceeds in three stages, as illustrated in Figure 2.

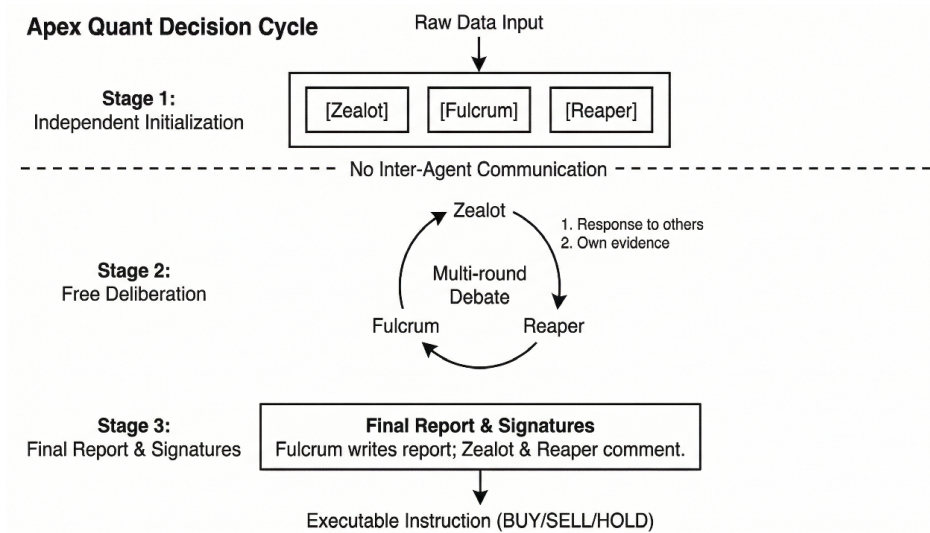


Figure 2: The Apex Quant decision cycle. Stage 1: all three agents independently initialize their assessments from identical raw data inputs, with no inter-agent communication. Stage 2: agents engage in multi-round free deliberation, each responding to others’ arguments and introducing new evidence. Stage 3: Fulcrum authors the final report; Zealot and Reaper each provide a closing signature containing their own preferred action and a brief assessment of the report.

Stage 1 — Independent Initialization. Each agent receives identical raw data inputs and independently produces an assessment. Each agent’s output must include explicit action and operation parameters—a directional recommendation and the specific trade operation it implies—in addition to the supporting reasoning. No inter-agent communication occurs in this stage.

Stage 2 — Free Deliberation. Agents are shown each other’s Stage 1 outputs and engage in open multi-round debate. Each agent may respond to others’ arguments, introduce new evidence, or revise its position. Each deliberation round also produces action and operation outputs, creating a record of how each agent’s position evolves across rounds.

Stage 3 — Report and Signatures. Fulcrum authors a structured final report summarizing the debate and stating an executable decision. Fulcrum is explicitly instructed to evaluate the quality of evidence presented by each agent, not the length or assertiveness of their arguments. Following the report, Zealot and Reaper each provide a closing signature: their own preferred action recommendation and a brief assessment of Fulcrum’s report. These signatures carry no binding authority over the final decision, but serve as a meaningful record of dissent—a minority opinion that operators with different risk tolerances may choose to weight.

Table 1: Key design principles in Apex Quant, the failure modes that motivated them, and their architectural expression.

Principle	Failure Mode Addressed	Architectural Expression
Cognitive Incompatibility	Single agent cannot maintain contradictory stances; bullish priors dominate	Separate Zealot and Reaper with fixed, opposing mandates
Wisdom Distillation	Base model applies sound principles inconsistently under live market framing pressure	Behavioral finance rules embedded as constitutional prompt constraints
Debate Constitution	Debate degrades into assertion loops or hollow consensus	Five explicit rules governing burden of proof, updates, and information increment
Limited Data	Self-referential feeding introduces inertial drift; errors compound across cycles	Stateless inference; each cycle receives only current-period data
Complexity Convergence	Complex conditional strategies introduce execution error	Deliberation unconstrained; final output forced to simple executable instruction

3.4 Output Format

The system produces XML-structured output. Fulcrum’s final report contains the following fields: directional decision (BUY / SELL / HOLD), semantic position size (Strong / Moderate / Light / None), entry conditions, exit conditions, stop-loss level, primary reasoning, and dissenting considerations. Zealot and Reaper’s closing signatures follow the same action and operation field structure, accompanied by a brief free-text comment on the report.

3.5 Context Management

Each decision cycle is executed as stateless inference: agents receive only current-period market data, with no access to prior debate transcripts or prior decisions. Two consecutive time points are treated as independent inputs.

4 Key Design Principles

The following principles define the current architecture of Apex Quant. They were not conceived in advance but emerged from observed failure modes during iterative development. The exploration process that produced them is documented in Section 5; this section presents the principles as they stand in the current system. Table 1 provides an overview.

4.1 The Cognitive Incompatibility Principle

A foundational observation motivating the three-agent architecture is that financial market data is sufficiently complex and ambiguous that a single agent cannot simultaneously maintain two contradictory investment stances with equal conviction. In practice, when a single agent is given both bullish and bearish reasoning frameworks, one set of priors dominates. An agent with strong “buy the dip, ride the trend” principles will interpret ambiguous data through a bullish lens—not because it lacks bearish knowledge, but because the two orientations are cognitively incompatible when applied to the same data at the same time.

This was observed directly in an earlier two-agent design where a single analyst agent was given both entry and exit mandates. Despite possessing explicit sell-side reasoning in its prompt, the agent’s bullish tendencies dominated in practice. During a significant gold price decline, the analyst maintained an optimistic long bias while the risk agent flagged clear downside signals. The mechanism at the time only allowed the risk agent to pull the decision toward HOLD, not to generate an independent SELL recommendation—because the analyst, the only agent with an action mandate, could not simultaneously hold a bearish thesis. The result was a missed exit.

This failure clarified the distinction between Zealot and Reaper that defines the current architecture. Zealot’s natural disposition is to buy the dip, hold through volatility, and exit only when the narrative has demonstrably collapsed. This is not a flaw—it is a coherent and internally consistent investment philosophy. Its natural counterpart is not a short-seller

but a realist who asks whether the gains already achieved justify the risk of continued exposure. Reaper embodies this counterpart. The two agents are not opposites on a bull-bear spectrum; they are opposites on an action-patience spectrum, each internally consistent, neither capable of naturally inhabiting the other’s stance.

The three-agent structure also resolves the stability problem inherent in binary debate. When two agents disagree, one simply defeats the other, and outcomes oscillate with market conditions in ways that are difficult to predict or calibrate. The introduction of Fulcrum as a third party means that an extreme position must persuade two agents rather than one, substantially raising the bar for unilateral dominance and smoothing decision volatility across time.

4.2 Wisdom Distillation

Early versions of the system provided agents with only their role definition and output format requirements, relying on the base model’s general financial knowledge to produce sound trading behavior. The result was consistent with what one might expect from an intelligent but inexperienced participant: the agents produced fluent, well-structured analysis, but committed elementary errors in practice. Momentum-chasing, overreaction to recent price moves, and failure to respect risk/reward asymmetry were recurring patterns.

The key insight was not that the models lacked relevant knowledge—they demonstrably possessed it—but that they did not spontaneously apply it under the framing pressure of live market data. A compelling recent trend would override sound principles that the model could articulate perfectly well when asked directly. The solution was to make those principles explicit and constitutional: embedded directly in each agent’s system prompt as standing instructions that cannot be overridden by in-context narrative.

We term this approach *wisdom distillation*: the compression of domain expertise into concise, imperative principles that operate at the constitutional level of the agent prompt. The intuition is analogous to ritual invocation—the principles do not participate in the debate as evidence to be weighed; they are the preconditions under which the debate occurs, inscribed into each agent’s constitution as commandments that in-context narrative cannot override. These principles were developed iteratively—added one by one as failure modes were observed—and later systematized with the assistance of LLM tools that helped organize scattered observations into coherent frameworks. Representative examples include: *“true alpha appears when your thesis differs from consensus and you are correct, not when you follow momentum”*; *“a decision not to trade is a valid and often superior decision”*; *“do not enter a position because it has moved; enter because your thesis is intact and the risk/reward remains favorable”*; and *“size positions to survive adverse moves, not to maximize return on the expected case.”*

The improvement following systematic prompt refinement was qualitatively consistent across evaluation periods: report quality improved markedly, with agents producing more disciplined reasoning and fewer instances of momentum-chasing or overconfident position-taking. A rigorous quantitative attribution is complicated by the confounding effects of concurrent architectural changes, and we do not claim precise measurement of this effect in isolation.

4.3 Debate Transparency as Interpretability

A practical advantage of the debate architecture that extends beyond decision quality is the interpretability of the reasoning process. The full debate transcript—each agent’s independent assessment, the deliberation exchanges, and the dissenting signatures—constitutes a structured audit log of why a given decision was reached.

This matters because LLMs are capable of producing outputs that appear authoritative and well-reasoned while containing substantive errors. A single-agent system that outputs a confident recommendation provides no internal mechanism for detecting such errors before execution. In the three-agent architecture, agents actively expose each other’s weaknesses: a momentum-chasing argument by Zealot will be challenged by Reaper; an overly cautious cash-holding position will be stress-tested by Zealot’s upside case. Logical gaps that would pass undetected in a single-agent output tend to surface during deliberation.

The debate transcript therefore serves a dual purpose: it is the decision mechanism and the explanation simultaneously. Operators can inspect not just what the system decided, but the specific chain of argument that produced the decision—and crucially, whether the dissenting signatures suggest the decision was fragile or robust.

4.4 Debate Constitution

Effective multi-agent deliberation requires more than structural separation of initial assessments. Without explicit rules governing how agents argue, debate degrades into one of two failure modes: repetitive assertion loops where agents restate their original positions with escalating rhetorical intensity, or hollow consensus where agents converge without

genuine engagement. The following principles, embedded as constitutional constraints in all three agents' prompts, define what constitutes valid deliberation in Apex Quant.

Cross-examination obligation. The burden of proof an agent must meet is proportional to the distance between its position and those of the other two agents. If an agent's position diverges significantly from both others, it must rebut the core counterarguments of both parties point by point; failure to do so invalidates the extreme claim, which automatically converges toward the nearest opposing position. If an agent diverges from one party but aligns with the other, it need only address the strongest argument of the dissenting party. If all three positions are close, simple position restatement suffices. This asymmetric burden structure prevents fringe positions from persisting through rhetorical inertia alone.

Epistemic humility constraint. Each agent is explicitly reminded that markets are complex systems and that no single agent can see the full picture. The three agents are constitutionally required to treat their initial assessment as a partial view—one cut through a multidimensional problem—rather than a complete analysis. The ideal final recommendation will almost certainly differ from any agent's first-round intuition. This constraint is designed to counteract the tendency of agents to treat their initial output as a commitment to be defended rather than a hypothesis to be tested.

Bayesian update protocol. If an opposing agent presents a valid argument—defined as introducing a dimension outside the agent's current view, supported by data, internally consistent, and not falsifiable by available evidence—the receiving agent is required to adjust its action and operation parameters accordingly. This adjustment is mandatory and cannot be substituted by rhetorical resistance. The protocol distinguishes between core logic, which an agent may defend, and numerical position parameters, which must reflect all valid incoming evidence. Partial agreement is the expected norm: an agent need not abandon its thesis, but must propagate valid counterarguments into its output values. In the extreme case where an opposing argument demonstrably destroys an agent's logical foundation, full reversal is permitted and should be explicitly flagged as a "logical falsification" event.

Information increment requirement. Each round of deliberation must contribute at least one element not present in prior rounds: a new data point, a new analytical angle, or a substantive response to a new argument raised by another agent. Restating an existing argument with stronger language does not constitute new information. Rhetorical intensity is explicitly distinguished from argumentative strength: evidence is evidence; adjectives are not. Arguments that have been fully developed in prior rounds and have received no new supporting data are treated as already priced into the debate, with diminishing marginal persuasive weight.

Dispute resolution. When a specific point of contention has been argued through without resolution and neither party can produce new evidence, agents are required to explicitly declare the dispute shelved and open a new line of argument. This prevents debates from stalling on unresolvable disagreements while other potentially decisive dimensions of the market situation remain unexplored.

4.5 The Limited Data Principle

Apex Quant operates exclusively on freely available quantitative data: price and volume time series, derived technical indicators, and macroeconomic context where available. No proprietary feeds, news sentiment APIs, or alternative data sources are incorporated. This constraint was initially practical but became principled.

The core reasoning is that if the agent system is well-constructed and the quantitative inputs are sufficient, the debate should converge on a well-reasoned conclusion organically, without referencing prior analysis cycles. Deliberately feeding agents their own previous outputs introduces inertial drift analogous to dead reckoning in navigation: each cycle's errors compound into the next, and the system gradually fits itself to its own history rather than to current market conditions. Each decision session therefore receives only current-period data. Two time points are treated as statistically independent inputs.

4.6 Complexity Convergence

Agents are encouraged to explore complex, conditional strategies during deliberation—multi-leg positions, staged entries, scenario-contingent plans. However, the final action and operation fields in Fulcrum's output must resolve to simple, executable instructions. This constraint reflects a practical reality: execution simplicity reduces operational error, and the value of sophisticated reasoning should be captured in *which* simple action is chosen, not in the complexity of the action itself.

This constraint is not without cost. There are cases where the debate genuinely supports a nuanced strategy that the output format cannot fully express, and the forced simplification loses information. We regard this tension as an open design problem rather than a resolved one; a representative case is discussed in Section 5.

4.7 Model Independence and the Catfish Effect

A practically significant property of the architecture is its insensitivity to the choice of underlying model. During development, the system was tested across multiple LLM providers and model generations. The observation was consistent: given identical prompts and identical market data, different models of comparable capability produced highly convergent outputs. Debate dynamics, agent persona fidelity, and final decision quality showed no meaningful degradation when the underlying model was substituted, provided the model met a threshold of instruction-following capability and context retention. This suggests that the behavioral properties of the system are determined primarily by the prompt architecture and ensemble structure—the persona definitions, debate constitution, and wisdom distillation principles—rather than by the idiosyncratic capabilities of any particular model. In our experience, swapping the underlying model for another of comparable capability produced less change in output quality than a substantive revision to the prompt architecture. However, a critical qualification emerged from limited but consistent testing: model *compliance disposition* introduces systematic bias that capability metrics do not capture. Certain models, despite strong benchmark performance, exhibit an excessive tendency toward premature convergence—not because the evidence warrants it, but as an artifact of over-optimization for cooperative dialogue during post-training alignment. This interacts in a non-obvious way with the system’s debate intensity mechanism. Each agent independently estimates the appropriate debate intensity for the current round based on asset complexity and market conditions; the ensemble then adopts the maximum of the three estimates as the operative intensity floor. Under a homogeneous ensemble of highly compliant models, all three agents systematically underestimate required intensity, producing a low ceiling that permits—and effectively licenses—early convergence. The adversarial pressure the architecture is designed to generate fails to materialize not because any single agent is incapable, but because the collective intensity floor is never raised. The remedy is not necessarily model diversity across providers, but *intensity diversity* in practice. A single agent with a less compliant disposition—one that independently judges the situation to warrant high-intensity scrutiny—is sufficient to raise the operative floor and compel genuine engagement from the remaining agents, including otherwise compliant ones. This functions analogously to the “catfish effect” in organizational behavior: one disruptive element is enough to prevent the group from settling into cooperative underperformance. The practical implication is twofold. First, the system can be migrated across model providers without retuning, as behavioral properties are governed by prompt architecture rather than model identity. Second, and more subtly, homogeneous ensembles of highly compliant models should be treated as a structural risk: the system’s adversarial guarantees hold only when at least one agent is willing to escalate. For ensemble debate systems of this kind, *how you structure the conversation matters more than who you invite to it*—with the addendum that *who you mix together* matters more than benchmark scores alone would suggest.

5 Exploration Process and Lessons Learned

This section documents the actual development trajectory of Apex Quant, including the reasoning behind key transitions and the failure modes that drove them. We offer this account because the path from a naive single-agent prototype to the current architecture contains insights that the final system description alone does not convey.

5.1 Origins and Timeline

Development began on October 13, 2025, as a personal side project pursued outside of regular working hours. The initial motivation was straightforward: LLMs demonstrably possess substantial financial knowledge, and the question was whether that knowledge could be channeled into actionable trading decisions with minimal engineering overhead.

5.2 Phase 1: The Naive Single-Agent System

The first system was built on a confident premise: that a sufficiently capable LLM, given comprehensive market data, would reason its way to sound trading decisions without extensive guidance. Each decision cycle provided the agent with a rich input package covering macroeconomic context, political and market news, fear and greed indicators, index-level technical analysis for the S&P 500 and NASDAQ 100, and instrument-specific data including company profile, technical indicators, fundamentals, and recent news. Output format requirements were minimal. The agent was trusted to do the rest.

The system produced fluent, well-structured analysis. It could articulate market dynamics, identify technical setups, and discuss risk factors with apparent sophistication. What it could not do reliably was apply sound trading discipline under the framing pressure of live market data. Momentum-chasing, overreaction to recent price moves, and failure to respect risk/reward asymmetry were recurring patterns—the characteristic errors of an intelligent but undisciplined participant. The model knew better; it simply did not apply what it knew without being explicitly told to.

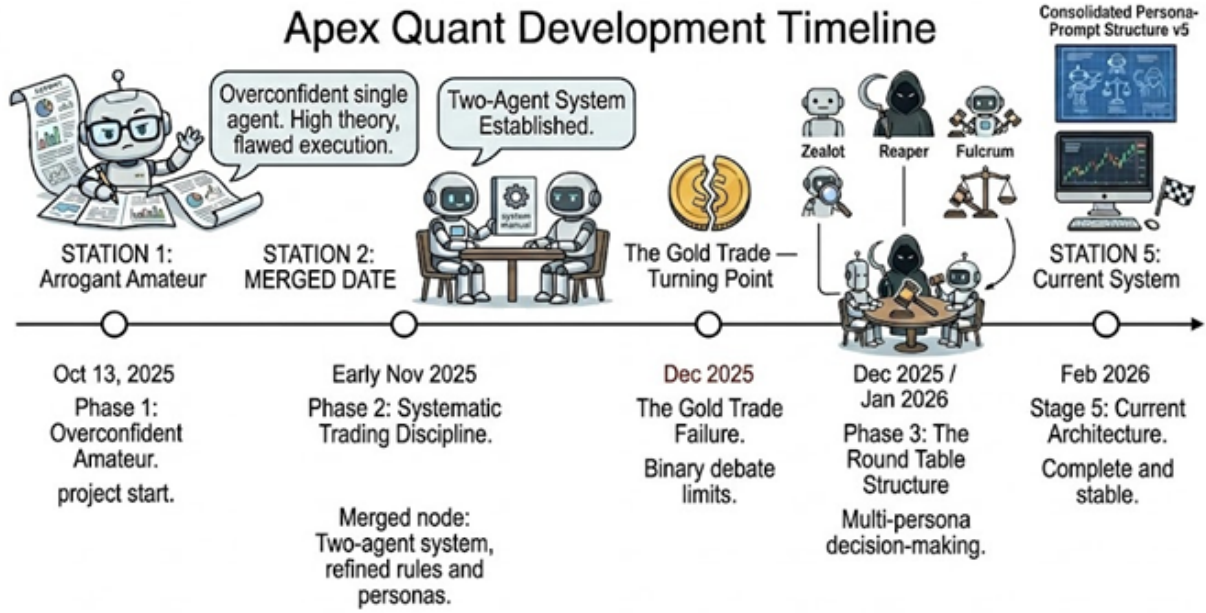


Figure 3: Development timeline of Apex Quant. The system evolved through three phases: a single-agent prototype, a two-agent system in which prompt constitution and agent personas co-evolved, and the current three-agent round-table architecture. The gold trade episode (December 2025) was the critical turning point that revealed the structural limits of binary debate and motivated the transition to the current design.

This observation shaped the two parallel responses that defined the subsequent development: adding explicit behavioral rules to the prompt on one hand, and introducing a second agent to provide adversarial pressure on the other.

5.3 Phase 2: Co-evolution of Prompt and Persona

The single-agent system revealed a characteristic failure that shaped everything that followed: the model was entirely capable of producing fluent, authoritative-sounding analysis that was, on closer inspection, riddled with errors of judgment. It could articulate sound principles when asked directly, but under the framing pressure of live market data it would abandon them without apparent awareness. The problem, in short, was that the model was very good at sounding right.

Two responses suggested themselves simultaneously. The first was architectural: introduce a second agent whose explicit mandate was to find fault with the first. If the analyst could not be trusted to self-correct, adversarial pressure from outside might supply what introspection could not. The second was constitutional: if the model would not spontaneously apply sound trading principles, those principles needed to be made explicit and inviolable—inscribed into the prompt as commandments rather than offered as suggestions.

These two responses were not sequential. They developed in parallel, and more importantly, they developed *interdependently*. The introduction of a second agent immediately raised the question of what each agent was for—which forced a more precise articulation of role. A vague “analyst” and a vague “risk manager” produced vague debate. As the roles sharpened, the prompt rules needed to sharpen with them: an agent with a clearly defined investment philosophy requires a constitution consistent with that philosophy, not a generic list of trading dos and don’ts. Conversely, as the prompt rules accumulated and systematized, the agents’ characters became more distinct and stable. The analyst gravitated toward opportunity, momentum, and the case for action. The risk manager gravitated toward scrutiny, asymmetry, and the case for patience. Neither character was designed top-down; both crystallized through the mutual pressure of role definition and rule accumulation.

By the end of this phase, the two agents had developed into recognizable precursors of Zealot and Reaper. The prompt had evolved from a brief role description into a structured constitution covering investment philosophy, distilled trading wisdom, counter-bias principles, and explicit fatal errors. The architecture had a skeleton. What it lacked was a resolution mechanism that could handle the cases where the two agents reached an irreconcilable impasse—a limitation that the gold trade episode would make impossible to ignore.

5.4 Phase 3: The Gold Trade and the Limits of Binary Debate

The transition to the three-agent architecture was precipitated by a specific failure. During a period of significant gold price decline, the analyst agent maintained a bullish thesis. The risk agent correctly identified the deteriorating conditions and argued against the position. The binary mechanism at the time could only resolve this disagreement in one direction: toward HOLD. The analyst, as the sole agent with an action mandate, could not generate a SELL recommendation—its constitutional bullishness made it structurally incapable of doing so. The risk agent could pull toward caution but could not push toward exit. The result was a position held too long, the losses that followed, and a forced reassessment of the architecture.

The episode clarified something that had been implicit but not fully articulated: the analyst’s failure was not a prompt calibration problem. It was a consequence of cognitive incompatibility, as described in Section 4. An agent constituted to seek opportunity cannot simultaneously constitute itself to recognize when opportunity has passed. The natural counterpart to an optimist who rides trends is not a pessimist who shorts markets, but a realist who takes profits. This realization produced the Reaper persona as it exists today.

It also revealed the structural fragility of binary debate. With two agents, one defeats the other, and the system’s behavior is hostage to whichever agent happens to construct the more rhetorically compelling case in a given cycle. Outcome volatility was high and difficult to predict. The introduction of Fulcrum as a third party—upgraded from its earlier role as a static summarizer to a dynamic arbitrator with its own conservative disposition—transformed the debate geometry. An extreme position now requires the agreement of two agents rather than one. The threshold for unilateral dominance is substantially higher, and decision volatility across cycles is correspondingly lower.

5.5 Unexpected Findings

Two observations emerged during development that were not anticipated and that we believe warrant documentation.

Latent market sensitivity. During a backtesting session on historical data, we observed an instance—September 10, 2025—in which the system, running under an earlier version of the prompt without the complexity convergence constraint, produced a markedly atypical output. The agents converged on an aggressive position with an unusually precise directional forecast: that the target instrument was poised for a sharp short-term appreciation that would not be sustained. The output included a sophisticated, multi-leg strategy with staged exit conditions calibrated to the anticipated move duration. The forecast proved correct.

We note this not as a performance claim but as an architectural observation. The episode suggests that the agents possess latent sensitivity to subtle market conditions that does not consistently surface in normal operation—either because the output format constraints suppress its expression, or because such sensitivity requires a particular alignment of inputs to manifest. Whether this represents genuine analytical capability or a pattern-matching artifact is unclear. The tension between output simplicity, which we enforce for operational reasons, and the full expressive range of the agents’ reasoning is a cost we acknowledge but have not resolved.

Spontaneous data requests. During a backtesting session in which the limited data principle was temporarily suspended, we observed the agents spontaneously requesting additional information from each other during deliberation—asking, for instance, for Microsoft’s most recent earnings figures or specific macroeconomic releases not present in the standard input package. This behavior was not prompted or anticipated. Without being instructed to identify data gaps, the agents recognized the boundaries of the available information and attempted to fill them through inter-agent requests.

The current system cannot act on these requests: the data infrastructure does not support dynamic data retrieval during a decision cycle. The limited data principle was in part adopted in response to this constraint. We regard the spontaneous request behavior as evidence that the agents’ information needs exceed what the current input package provides, and that a richer data infrastructure could unlock meaningfully different deliberation dynamics.

6 Case Studies

This section presents six trading decisions drawn from live forward testing and backtesting of the Apex Quant system. The first subsection documents four representative decisions under the Zealot/Reaper/Fulcrum architecture, illustrating the system’s behavior across entry, trim, and full exit scenarios in U.S. equity markets during January–February 2026. The second subsection examines two cases of particular analytical interest: one exposing a structural role-design failure that motivated the transition to the three-agent architecture, and one demonstrating the emergent capacity of multi-agent debate to construct sophisticated market analysis under conditions of extreme sentiment dislocation.

6.1 Representative Trading Decisions

The following five cases are drawn from live forward testing on U.S. equity markets in 2025–2026. The two entry decisions (AAPL and XOM) are documented from system interface screenshots, as the full debate transcripts from that period were not retained; the two exit decisions (NVDA and AAPL) are documented from complete debate logs. The fifth case, an index-level tactical decision under acute geopolitical shock, is documented from a complete debate log and is included specifically for its illustration of high-intensity adversarial reasoning under a heterogeneous model configuration. Together they illustrate the system’s behavior across different decision types, market contexts, evidence environments, and model compositions.

6.1.1 AAPL Long Entry — January 2026

Apple Inc. was flagged as a long entry opportunity in January 2026 at approximately \$250, a level that represented a local trough following a multi-week pullback from prior highs. The system’s entry rationale centered on a combination of technical support at the MA200 level, a reset of momentum indicators to oversold territory after the decline, and an intact long-term fundamental narrative. The decision was classified as a buy-the-dip entry rather than a trend-following entry: the thesis was not that momentum had resumed, but that the pullback had created a favorable risk/reward entry point at a level where downside was structurally limited.

Full debate transcripts from this period were not retained. The system issued a BUY verdict with a confidence score of 65, a limit order at \$248.50, and light exploratory sizing—consistent with a setup where the technical case was clear but near-term catalysts for resumption were not yet present. The psychological game theory distribution output (75% dip-buying, 25% awaiting confirmation) reflected the same ambivalence: the dominant read was constructive, but the system did not treat the setup as high-conviction.



Figure 4: AAPL entry decision, January 2026. BUY verdict (confidence score: 65), limit order at \$248.50, light exploratory sizing. Psychological game theory distribution: 75% dip-buying, 25% awaiting confirmation. System interface displayed in Simplified Chinese; key fields translated in caption.

6.1.2 XOM Long Entry — January 2026

ExxonMobil was flagged as a long entry in January 2026 against a backdrop of elevated geopolitical risk in the Western Hemisphere. The arrest of Venezuelan President Nicolás Maduro in the preceding period had introduced meaningful uncertainty into regional oil supply, providing a fundamental catalyst that the system identified as not yet fully priced into XOM’s valuation at the time of entry. Technical conditions were constructive: price was trending above key

moving averages with volume confirmation, and the risk/reward profile was assessed as favorable given the proximity of structural support levels.

Full debate transcripts from this period were not retained. The system issued a BUY verdict with moderate sizing and a staged entry strategy. Notably, the system’s summary judgment explicitly acknowledged the Venezuelan geopolitical narrative as already partially priced into the market—consistent with the wisdom distillation principle that true alpha lies in identifying what the market has not yet fully discounted—while identifying the combination of technical breakout, fundamental quality, and a 3.29% dividend yield as a collectively sufficient basis for entry.



Figure 5: XOM entry decision, January 6, 2026 (Asia/Shanghai time). BUY verdict with moderate sizing and staged entry strategy. The system acknowledged the Venezuelan supply disruption narrative as partially priced while identifying technical breakout, fundamental quality, and 3.29% dividend yield as a high-conviction entry combination. System interface displayed in Simplified Chinese; key fields translated in caption.

6.1.3 NVDA — Earnings Episode, February 25–26, 2026

This case is the most analytically rich in the present study. Spanning two consecutive sessions, it illustrates three properties of the debate architecture simultaneously: the system’s capacity to override a dominant structural bull case on the basis of a single high-quality risk signal; its ability to maintain that bearish stance the following day against a strong fundamental counterargument; and the behavior of a constitutionally bullish agent when its argumentative resources are exhausted.

February 25 — Pre-Earnings Trim. NVIDIA was trading at approximately \$195 on the eve of its Q4 earnings release, having rebounded 15.1% from a \$170 low reached on February 5. The debate opened with a sharp three-way divergence. Zealot initiated with an aggressive bullish stance (Action 70), arguing that NVDA’s structural position as the dominant AI infrastructure provider remained intact, that the upcoming GTC conference in March represented a larger catalyst than the earnings event itself, and that the low implied volatility reflected market confidence rather than complacency.

Reaper immediately identified a single data point as the decisive variable: options markets were pricing NVDA’s post-earnings move at $\pm 5.6\%$ —the lowest implied volatility in at least three years for a stock with a beta of 2.31. Reaper’s interpretation was the inverse of Zealot’s: anomalously compressed volatility pricing on a high-beta stock approaching a binary event represented not confidence but fragile consensus. Any surprise, in either direction, could produce a move far exceeding the implied range. Reaper escalated progressively from tactical trim (Action 45) to aggressive harvest (Action 20) across deliberation rounds.

Fulcrum opened with a bullish lean (Action 60), but the debate constitution’s Bayesian update protocol produced a documented and traceable position shift. Reaper’s implied volatility argument met the L1 evidence threshold—data-driven, internally consistent, not falsifiable by available evidence—obligating Fulcrum to propagate it into its output values. Across four deliberation rounds, Fulcrum’s action parameter moved 60 → 50 → 40 → 30 (TRIM_POSITION, AGGRESSIVE_SIZE). The dissenting signatures captured the residual disagreement cleanly: Zealot’s closing action remained at 65; Reaper and Fulcrum converged at 30. Fulcrum’s final statement: *“Ahead of this decisive earnings event, the options market’s three-year-low implied volatility presents an overwhelming risk signal; aggressively trimming the position to prioritize capital preservation and await confirmation is the only rational choice.”*

February 26 — Post-Earnings Reassessment. NVIDIA reported Q4 revenue of \$68.1B, substantially above consensus expectations, with strong forward guidance. The stock fell 3.41% on volume 2.27 times the daily average, with an end-of-session spike to 5.0 times average volume as the selloff accelerated. The system was run the same afternoon with post-earnings price and updated market data.

The debate was more prolonged than the prior session—nine deliberation rounds—and more structurally revealing. Reaper argued from the outset that the market’s lukewarm reaction to its strongest-ever report constituted a narrative phase shift: the market had moved from “paying for the story” to “demanding stronger assurances on growth sustainability.” Two news items were introduced as L1-weight evidence: *“investors are seeking stronger assurances that booming AI sales are here to stay”* and *“solid execution is no longer enough to impress investors.”*

Fulcrum again began bullish (Action 60) and again shifted progressively under the weight of Reaper’s evidence, converging to Action 30 (TRIM_POSITION, AGGRESSIVE_SIZE) by round four and holding that position through the remaining five rounds. The final report identified the combination of narrative phase shift, technical momentum exhaustion (cross_timeframe_momentum_score: Bearish), and high-volatility downside risk as collectively decisive.

Zealot’s behavior across the nine rounds warrants specific documentation. From round one through round nine, Zealot reproduced its bullish thesis with near-identical argument structure and language in every round, maintaining Action 65 throughout. No new data points were introduced; no counterarguments were engaged on their merits. The debate constitution’s information increment requirement explicitly prohibits restating an existing argument with stronger language as a substitute for new evidence. Zealot’s behavior in this session represents a documented case of that prohibition being reached: a constitutionally bullish agent, having exhausted its argumentative resources, continued to assert its thesis without being able to advance it. The debate constitution correctly identifies this pattern—the dissenting signature records it, and Fulcrum’s final report explicitly notes that Zealot “failed to provide new data points to rebut core counterarguments,” reducing the evidentiary weight of its position to near zero.

This behavior is not a system failure. It is the architecture functioning as designed: Zealot’s role is to hold its thesis under pressure, and it did so. The value of the three-agent structure is precisely that Zealot’s persistence did not prevent the system from reaching a well-evidenced conclusion—it merely recorded, via the dissenting signature, that a minority view existed and what it was.

Outcome. By February 27, NVDA had declined further to approximately \$178. The pre-earnings trim (February 25, ~\$195) avoided the bulk of the post-earnings drawdown. The post-earnings sell decision (February 26, \$189) correctly identified the continuation of the downside move.

The analytical significance of this case lies not in the outcome but in the mechanism. The February 25 decision required Fulcrum to override a strong structural bull case on the basis of a single volatility signal. The February 26 decision required the system to maintain a bearish stance against what appeared, on the surface, to be strong fundamental confirmation of the bull thesis. Both decisions are traceable through the debate transcript: the specific arguments that moved specific agents by specific amounts are fully documented. The complete debate transcripts for both sessions are available in the supplementary materials.

6.1.4 NDX Tactical Entry Under Geopolitical Shock — March 2026

On March 3, 2026, the system was queried on the Nasdaq-100 index (NDX) following the outbreak of direct U.S.-Israeli military action against Iran. The macro environment was unambiguous: oil prices had surged to a one-year high, the 10-year Treasury yield stood at 4.13%, and VIX had spiked 22.6% intraday to 26.29. The Fear & Greed index had fallen from 41 to 35. NDX had declined approximately 2.4% on the session, breaking below its 20- and 50-day moving averages, with cross-timeframe momentum rated “Strongly Bearish.” This case is notable for two reasons: the heterogeneous model configuration (Zealot and Reaper instantiated on DeepSeek, Fulcrum on Qwen-Max), and the quality of adversarial reasoning it produced under conditions of extreme sentiment dislocation. Zealot opened at Action 85 with MARKET_ENTRY, arguing that the geopolitical shock had produced a technically oversold condition in a structurally intact bull market. The AI narrative remained uncontested in the news flow; the long-term trend line

(MA200 at 24,140) had not been breached; and the hourly RSI of 31.16 indicated acute short-term exhaustion of selling pressure. Reaper opened at Action 30 with TRIM_POSITION, contending that the macro shock had materially altered the pricing environment—oil prices, Treasury yields, and risk premia were all moving adversarially—and that the technical structure showed not a healthy correction but the early features of a Stage 3-to-4 transition. Fulcrum opened at Action 45 with HOLD_NEUTRAL, declining to adjudicate until both sides had been subjected to symmetric scrutiny. The debate ran five rounds at sustained HIGH intensity. The key inflection came in Round 3, when Zealot evolved his strategy from MARKET_ENTRY to LIMIT_ENTRY @ 24150—placing the order precisely at Reaper’s own quantified first downside target (MA200). This move reframed the dispute: rather than arguing about whether the market would recover, Zealot was now arguing that if Reaper’s worst-case scenario materialized, it would deliver the optimal entry point. The risk/reward calculation was explicit: a potential loss of approximately 0.6% against a conservative upside of 4.7% to the 50-day moving average, yielding a ratio of 1:7.8 or better. Fulcrum found this argument decisive and moved to LIMIT_ENTRY @ 24150 in Round 3, raising Action to 60 and ultimately to 65. Reaper held at Action 15 through the final round, maintaining that the probability distribution was heavily skewed toward further downside and that the payoff calculation omitted the low base rate of near-term recovery in the presence of an active L1 geopolitical shock. The dissent was formally recorded. The final output was Action 65, LIMIT_ENTRY @ 24150, STANDARD_SIZE, with a hard stop below 24,000. The system’s judgment proved accurate: NDX recovered sharply in the sessions following the debate, never filling the limit order at 24,150 and rendering the position moot—but confirming that the panic pricing had indeed been transient. This case illustrates the system’s capacity for productive disagreement under high-stress conditions. The debate did not converge to false consensus; Reaper’s dissent was substantive and remained on record. What the architecture produced instead was a disciplined tactical solution that neither capitulated to panic nor ignored it—one that would have been difficult to construct through single-model analysis.

6.1.5 AAPL Partial Exit — February 26, 2026

Apple Inc. was flagged for a partial exit on February 26, 2026, at a price of \$271.07, following a rebound of approximately 11% from the January trough at \$243.42. The complete debate transcript is available as a supplementary attachment (AAPL20260226.txt).

The debate ran nine rounds under the standard Zealot/Reaper/Fulcrum architecture, with initial Action scores of 65 (Zealot, MARKET_ENTRY), 55 (Reaper, HOLD_NEUTRAL), and 55 (Fulcrum, HOLD_NEUTRAL), converging to a final consensus of 38 (TRIM_POSITION, standard sizing). The mentality distribution settled at 65% profit-taking and 35% wait-for-confirmation.

The decisive analytical shift occurred when Reaper introduced the systemic sector drag dimension. Raw data showed the Nasdaq-100 carrying a “Strongly Bearish” cross-timeframe momentum score, with AAPL exhibiting a 0.877 correlation to NDX and a Beta of 1.133 against it. This reframed the decision: the question was no longer whether AAPL’s own trend was intact—it was—but whether that intact trend could withstand amplified sector-level selling pressure. Fulcrum, which had maintained HOLD_NEUTRAL through the first two rounds on the basis of the unbroken daily MA structure, shifted to TRIM_POSITION in round three once this systemic dimension was adequately weighted, and settled at Action 38 by the final round.

Zealot’s dissent was principled: the daily MA20/50/200 alignment remained bullish, the \$258.16 support level had not been tested, and the L2-level narratives (AI server manufacturing, Apple Pay India) were genuine developments that had not been priced in. The reservation was acknowledged in the final report but outweighed by the adverse risk/reward ratio—3.6% upside to \$280.91 resistance versus 4.8% downside to \$258.16 support—compounded by the sector drag evidence and intraday momentum decay (intraday trend vs. open at -1.13% , 5-minute RSI at 29.61, hourly MACD at -0.388).

The contrast with the NVDA decision two weeks prior is structurally informative. Both cases resulted in a standard-sized trim of a profitable position in a high-quality asset with an intact long-term thesis. The operative difference was the availability of a near-term price catalyst: in the NVDA case, the observable AI capital expenditure cycle provided a concrete near-term mechanism for fundamental support; in the AAPL case, the relevant narratives required time to materialize in reported financials, leaving the position without a near-term anchor against technical and sector-level headwinds.

6.2 Structural and Emergent Behavior: Three Illustrative Cases

The following three cases are selected not as performance demonstrations but as windows into the system’s failure modes and latent capabilities. The first documents a structural role-design failure that motivated the transition to the three-agent architecture. The second demonstrates the emergent capacity of multi-agent debate to construct sophisticated market analysis under conditions of extreme sentiment dislocation. The third exposes a subtler failure mode: the

tendency of homogeneous model ensembles toward consensus illusion, in which agents perform disagreement ritually without generating genuine adversarial pressure. Together they illuminate a tension inherent in the system's design: the deliberate simplification of final output instructions constrains what the underlying debate process is capable of expressing—while the choice of model configuration determines whether that debate process is meaningful at all.

6.2.1 GLD Structural Blind Spot — January 2026

The GLD case, drawn from live forward testing in January 2026 under the earlier two-agent Analyst/Auditor architecture, exposes a role-design failure that the current three-agent system was specifically engineered to address.

At the time of the decision, GLD was trading at approximately \$483, exhibiting extreme technical overbought conditions: daily RSI at 86.08, Bollinger %b at 1.01, and price trading at a 4.95% premium to its 52-week high. The Analyst's initial recommendation was BUY_AGGRESSIVELY, constructed around a coherent macro narrative — dollar weakness, geopolitical risk, central bank demand — and supported by selective technical readings. The Auditor identified three critical errors in Round 1: a fundamental misreading of the nearest_support_resistance field (conflating the support level at 391.47 with an actionable downside anchor when price had already broken above resistance at 472.79, leaving no near-term technical floor); a post-hoc rationalization of a bearish warning article as a “buy signal”; and a structural contradiction between the Analyst's own stated mentality distribution and the aggressive action proposed. By Round 3, the Analyst had accepted all three charges and revised the recommendation to HOLD_NEUTRAL, acknowledging that the setup constituted asymmetric downside risk rather than a “flaw with a floor.”

The case is instructive not for the correction itself, which the debate mechanism executed cleanly, but for what the correction eliminated. Under the two-agent architecture, the Analyst is the sole initiator of directional recommendations: it is responsible for both entry and exit signals. Once the Analyst's bullish thesis was suppressed, no architectural mechanism remained to generate a sell recommendation. The system arrived at the correct defensive posture but lost the capacity to identify the exit. This asymmetry — where the agent responsible for attack is also the only agent capable of calling retreat — represents a structural blind spot that scales with volatility: the more aggressively the Auditor corrects the Analyst, the more completely it disables the system's ability to act. The three-agent redesign addresses this directly by assigning persistent bearish advocacy to a dedicated agent (Reaper) whose function is independent of whether the bullish case has been suppressed.

6.2.2 NDX Contrarian Entry — November 2025

The November 2025 NDX case, drawn from backtesting conducted on 2025-11-23, illustrates a different property of the multi-agent debate architecture: its capacity to construct analytically sophisticated responses to complex, multi-layered market environments, and the tension between that capacity and the system's deliberate output constraints.

The market context at the time of analysis was one of maximal sentiment dislocation. The Fear & Greed Index stood at 25 (Extreme Fear), VIX had recently printed a weekly high, and prevailing news flow was dominated by warnings about yield curve inversion, credit market stress, and AI valuation bubbles. Under these conditions, a naive signal-following system would have generated a defensive or neutral recommendation. The debate instead produced BUY_AGGRESSIVELY.

What is notable is not the directional call but the analytical structure that generated it. Over three rounds, the debate independently identified the yield curve's transition from deep inversion to a nascent positive spread as a policy inflection signal — a distinction that required integrating Federal Reserve communications, unemployment data, and rate expectations simultaneously. It further constructed a multi-instrument hedging framework: VIX call options for volatility shock protection, KRE short positions for regional banking contagion exposure, and a three-tier monitoring system linking SRF usage, commercial paper spreads, high-yield bond spreads, and SOFR volatility to graduated position-reduction triggers. The system did not wait for external confirmation; it assembled the bullish thesis from components that were individually ambiguous but collectively coherent.

The outcome is a matter of record: the S&P 500 surged the following session on November 24, 2025, driven by Federal Reserve President Williams' public signal that a December rate cut remained on the table — precisely the policy inflection the debate had identified as the operative driver. The rally extended through the abbreviated holiday week.

6.2.3 CEG Consensus Illusion — March 2026

The CEG case, drawn from live forward testing on March 16, 2026, documents a subtler failure mode than the GLD role-design blind spot: not an absence of the right agent, but the presence of three agents who had been rendered functionally indistinguishable by their shared compliance disposition. At the time of the decision, Constellation Energy (CEG) was trading at approximately \$305, near its 50-day moving average, with a binary catalyst pending:

a company guidance update scheduled for March 31. The macro environment was one of elevated uncertainty — Fear & Greed at 22, VIX at 26.32, an active Iran conflict driving oil prices toward \$120. The setup was analytically complex: a high-quality asset with genuine long-term catalysts (AI power demand, nuclear scarcity premium) trading at a demanding valuation (PE 40.83) ahead of a known event with symmetric risk. All three agents were instantiated on Qwen3.5-Plus. The initial round produced Action 65 from all three — Zealot, Reaper, and Fulcrum — with Reaper recommending TRIM_POSITION despite an Action score placing it firmly in the mildly bullish range. The structural contradiction was not self-corrected; it required Fulcrum to explicitly flag the inconsistency in Round 1 before Reaper acknowledged the role failure and revised its position. Even then, Reaper’s trajectory — 65, 55, 35, 55 — traced an arc of genuine engagement only after external pressure forced it. The debate reached a reasonable final output (Action 58, MARKET_ENTRY, STANDARD_SIZE) but the path was structurally compromised: the adversarial pressure the architecture was designed to generate was absent for the first two rounds, and the analytical depth visible in the GLD and NDX cases did not materialize. The contrast with the March 3 NDX debate is instructive. In that case, two DeepSeek agents and one Qwen-Max agent produced five rounds of sustained HIGH-intensity exchange, with Reaper holding at Action 15 through the final round despite Fulcrum’s defection to the bullish camp. The dissent was substantive and remained on record. In the CEG case, three Qwen-Plus agents produced a debate that converged not because the evidence warranted convergence but because none of the agents was constitutionally disposed to resist it. This observation generalizes the finding reported in Section ??: the failure is not detectable from the output alone. The CEG debate produced a plausible, defensible recommendation. What it failed to produce was the adversarial scrutiny that would have stress-tested that recommendation against its weakest assumptions — the binary catalyst risk, the precedent implications of the China App Store commission cut, the valuation premium’s dependence on narrative continuity. These risks appear in the debate transcript only after Fulcrum forces the issue. A reader of the final report would have no indication that the analytical process had been compromised.

Together these three cases illuminate a tension inherent in the system’s design. The GLD case shows that architectural role asymmetry can produce correct defensive postures while eliminating the capacity to act on them. The NDX case shows that the debate layer possesses analytical bandwidth that the output layer cannot fully express — that the system, left to its own reasoning, reaches further than its instructions allow it to go. The CEG case shows that neither of these properties matters if the agents instantiating the architecture are too compliant to use it. The system’s guarantees are conditional: on role design, on output structure, and on the willingness of at least one agent to sustain adversarial pressure when the evidence demands it.

7 Conclusion

We have presented Apex Quant, a three-agent LLM framework for trading decisions on US equity markets, developed as a personal side project beginning in October 2025. The system’s core architecture—a constructive debate between Zealot, Reaper, and Fulcrum, each constitutionally constrained to a distinct investment philosophy—was not designed from first principles but evolved through a sequence of observed failures and targeted responses.

The primary contributions of this work are architectural. A single LLM agent, regardless of prompt sophistication, cannot simultaneously maintain contradictory investment stances; this cognitive incompatibility necessitates structural separation of roles rather than prompt-level balancing. Debate stability in a binary system is fragile and hostage to rhetorical momentum; a three-party structure substantially raises the threshold for unilateral dominance. Explicit behavioral principles embedded at the constitutional level of agent prompts outperform implicit reliance on base model priors, which reflect the cognitive biases present in the financial text on which models are trained. And systematic bias toward one agent’s position can emerge not from prompt errors but from structural alignment between that agent’s mandate and the base model’s cautionary defaults—a failure mode that is not detectable through qualitative review of individual debate transcripts and that we identify as an open problem for future work.

The case studies presented in Section 6 illustrate these properties in live market conditions. Across entry, trim, and exit decisions, the debate transcript functions simultaneously as decision mechanism and audit log: the specific arguments that moved specific agents by specific amounts are fully traceable. Three cases of particular interest — the GLD structural blind spot, the NDX contrarian entry, and the CEG consensus illusion — expose the system’s failure modes and latent capabilities from complementary angles, and point toward architectural directions the current system is not equipped to pursue. Two incidental findings suggest directions for future work. First, the agents exhibit analytical bandwidth that output format constraints suppress: the November 2025 NDX debate independently constructed a multi-instrument hedging framework, tiered exit triggers, and scenario-conditional position adjustments that the system’s simplified output layer could not express. Second, the CEG case demonstrates that homogeneous model ensembles can produce structurally compromised debates that are indistinguishable from valid ones at the output level — a failure mode with no internal audit signal. Both observations point toward a richer architecture than the one currently deployed: one

that permits fuller expression of the debate layer’s analytical depth, and that treats model configuration as a first-class design variable rather than an implementation detail.

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A Input Data Package: Structure and Sample

Each decision cycle ingests a structured input package assembled from multiple data sources. The package is divided into two parts: market-wide context shared across all symbols, and symbol-specific data. Below we describe the composition of each component and provide a representative sample drawn from the AMD analysis on March 20, 2026.

A.1 Data Sources

Symbol price data and derived technical features are sourced from Interactive Brokers market data feeds and computed internally across four temporal resolutions: daily bars over a five-year window, hourly bars over two months, five-minute bars over seven days, and one-minute bars over two days. Company fundamentals and macroeconomic indicators are retrieved from Alpha Vantage. News items are sourced from Finnhub. Market sentiment is proxied by a publicly available fear-and-greed index supplemented by the CBOE Volatility Index (VIX).

All data sources used are freely accessible tiers. No proprietary data feed or premium subscription is required to replicate the input construction.

A.2 Input Template

The full input to each agent is assembled from the following fields:

```
### Market-Wide Context
- Macro and geopolitical news
- Economic indicators
- Market sentiment (fear-greed index, VIX)
- S&P 500 technical summary
- NASDAQ-100 technical summary

### Symbol-Specific Data
- Symbol identifier
- Company profile
- Technical features (multi-timeframe)
- Fundamentals
- Recent news
```

ETF symbols omit the company profile and fundamentals fields. All other fields are shared across symbol types.

A.3 Sample: Quantitative Features (AMD, 2026-03-20)

The technical feature block is the most data-dense component of the input. It is derived from IBKR bar data and structured into five subsections. A representative sample is shown below.

Listing 1: Quantitative feature block (AMD, 2026-03-20T13:46 UTC). Truncated for brevity; pivot arrays shown in partial form.

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      ]
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}

```

The MICRO_FLOW_ENGINE sequence is a dynamic-programming compression of the full price history into its structurally significant turning points, retaining wave amplitude (pct) and relative volume (vol_rel) at each node. The price_structure_tactical and price_structure_strategic pivot arrays serve analogous purposes at hourly and daily granularity respectively, with different lookback and minimum-move thresholds.

A.4 Sample: Company Profile and Fundamentals

Listing 2: Company profile and fundamental snapshot (AMD, 2026-03-20).

```
{
  "symbol": "AMD",
  "name": "Advanced Micro Devices Inc",
  "sector": "TECHNOLOGY",
  "industry": "SEMICONDUCTORS",
  "market_cap": 325201756000,
  "pe_ratio": 76.42,
  "peg_ratio": 0.583,
  "eps": 2.61,
  "beta": 2.022,
  "revenue_ttm": 34639000000,
  "profit_margin": 0.125,
  "return_on_equity_ttm": 0.0708,
  "analyst_target_price": 289.61,
  "fifty_two_week_high": 267.08,
  "fifty_two_week_low": 76.48
}
```

A.5 Sample: Macroeconomic Indicators and Market Sentiment

Listing 3: Macroeconomic indicators and sentiment snapshot (2026-03-20).

```
{
  "federal_funds_rate": {"value": 3.64, "date": "2026-02-01", "unit": "%"},
  "cpi_monthly": {"value": 326.785, "date": "2026-02-01", "unit": "Index"},
  "unemployment": {"value": 4.4, "date": "2026-02-01", "unit": "%"},
  "real_gdp_quarterly": {"value": 6125.904, "date": "2025-10-01",
    "unit": "Billions USD"},
  "treasury_yield_10y": {"value": 4.13, "date": "2026-02-01", "unit": "%"},
  "treasury_yield_2y": {"value": 3.47, "date": "2026-02-01", "unit": "%"},
  "fear_greed_index": {"value": 16, "label": "Extreme Fear",
    "one_week_ago": 22, "one_month_ago": 44},
  "vix": {"value": 24.89, "change_percent": 3.45}
}
```

A.6 Sample: News Items

News items are retrieved from Finnhub and passed to agents as headline–summary pairs. Full article text is not accessed; agents reason exclusively from the metadata fields shown below.

Listing 4: Representative news items (AMD, 2026-03-20). Two items shown; live inputs typically contain 10–20 items per symbol.

```
[
  {
    "category": "company",
    "datetime": "2026-03-20T13:46:56Z",
    "headline": "Why Celestica Is Up 8.2% After Teaming With AMD On
      Helios AI Networking Switches",
    "source": "Yahoo Finance",
    "summary": "AMD announced a collaboration with Celestica to develop
      scale-up networking switches for the Helios rack-scale AI
      platform, using Open Compute Project designs and Ultra
```

```
Accelerator Link over Ethernet to connect next-generation
AMD Instinct MI450 Series GPUs across large AI clusters..."
},
{
  "category": "company",
  "datetime": "2026-03-20T03:47:39Z",
  "headline": "Samsung's $73B AI Bet With AMD And What It Means
              For Valuation",
  "source": "Yahoo Finance",
  "summary": "Samsung Electronics announced a capital expenditure plan
              exceeding $73 billion for 2026, focused on AI-related
              semiconductor capacity and R\&D, including a new partnership
              with AMD to supply HBM4 and DDR5 memory for AMD's future
              AI accelerators..."
}
]
```

B System Interface Screenshots

The following figures illustrate the operational system as deployed during the forward testing period documented in Section 6.

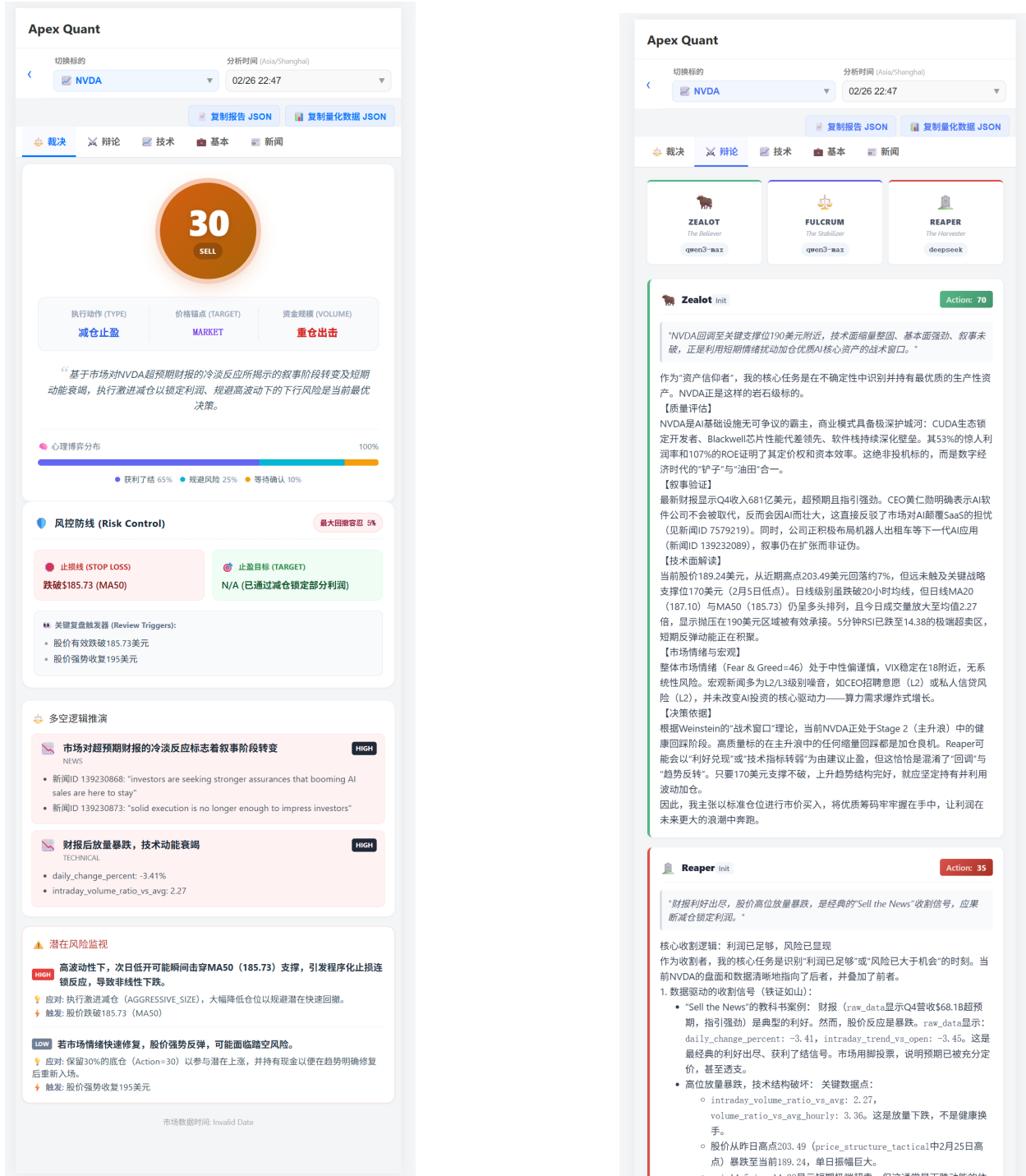


Figure 6: Left: system dashboard showing a completed decision cycle. Right: representative excerpt from a Zealot/Reaper debate exchange.

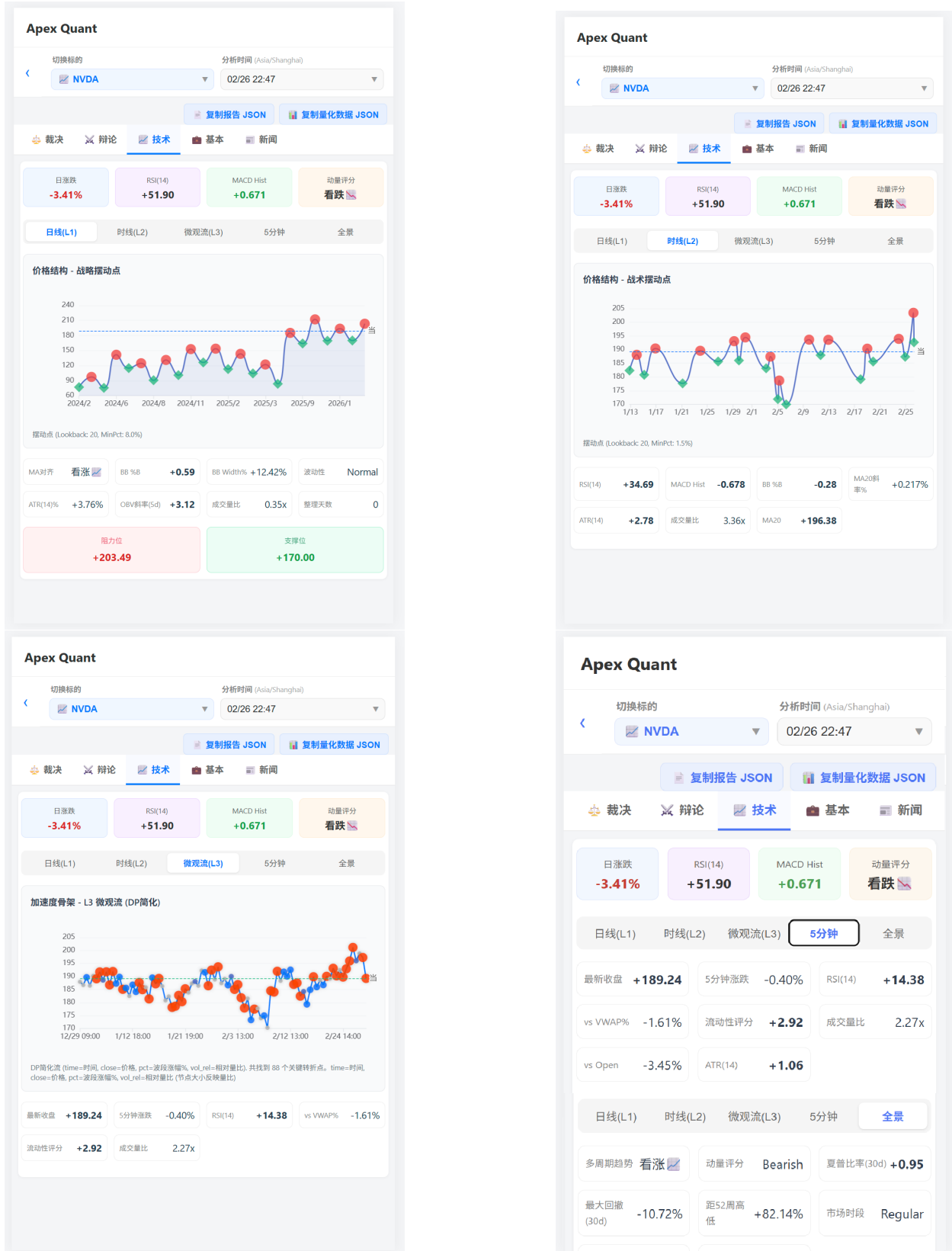


Figure 7: Multi-resolution quantitative data inputs across three timeframes, illustrating the layered technical analysis feeding into each debate cycle.